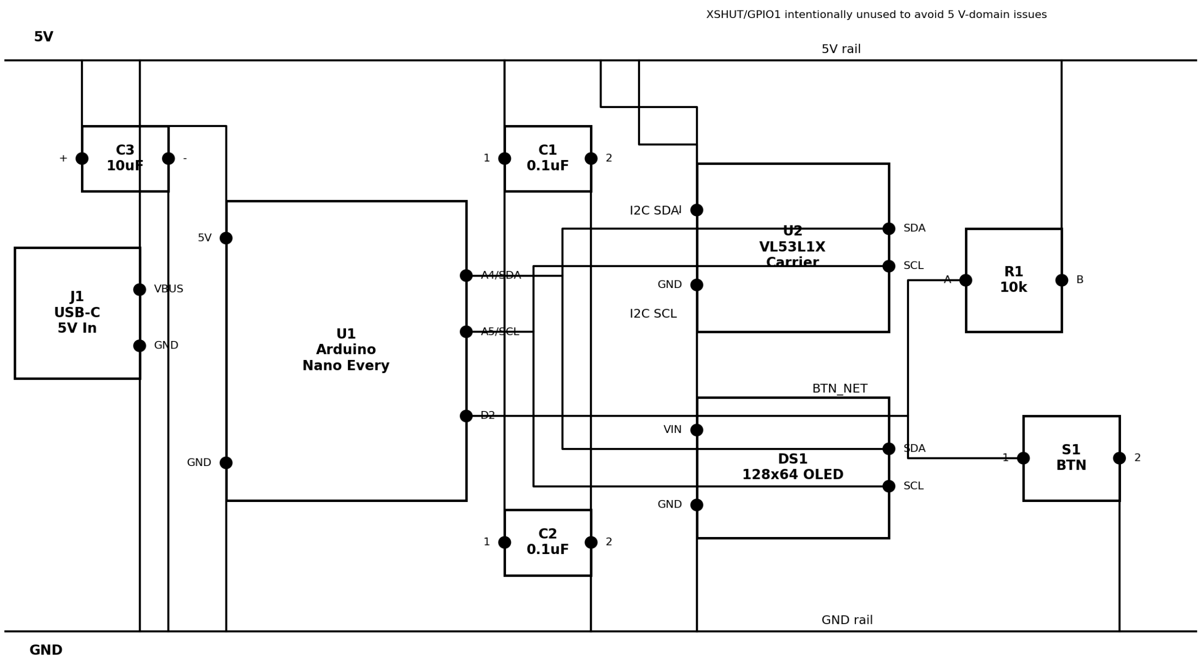
Rangefinder with OLED Screen and Trigger Button

Prototype design package for a USB-powered handheld distance meter using a VL53L1X time-of-flight sensor, Arduino Nano Every, 128x64 OLED display, and a single pushbutton to trigger measurements.

Assumptions: USB 5 V power, module-based build, no battery subsystem, and no use of VL53L1X XSHUT/GPIO1 to avoid 5 V logic-domain issues on the selected carrier board.

USB-Powered Rangefinder with OLED and Trigger Button



Key electrical summary

• Shared I2C bus: Nano A4/A5 connect to both the OLED and VL53L1X carrier.

• Button input is active low using an external 10 kΩ pull-up to +5 V.

• USB-C breakout provides the 5 V rail. Add local bulk and ceramic decoupling near the wiring cluster.

• The chosen VL53L1X carrier shifts SDA/SCL but not XSHUT/GPIO1; those pins are intentionally left unconnected in this design.

Bill of Materials

Item	Refs	Qty	Description	MPN	Unit $	Ext $	Supplier
1	U1	1	Arduino Nano Every	ABX00028	12.90	12.90	Arduino Store
2	U2	1	VL53L1X ToF sensor carrier	Pololu 3415	22.95	22.95	Pololu
3	DS1	1	0.96 in OLED display	Adafruit 326	17.50	17.50	Adafruit
4	S1	1	Momentary pushbutton	TEA-5144	0.89	0.89	Vetco
5	R1	1	Pull-up resistor	Generic 10k THT	0.10	0.10	Estimated
6	C1,C2	2	Ceramic decoupling capacitor	FA18X7R1H104KNU06	0.10	0.20	Futurlec/estimate

Item	Refs	Qty	Description	MPN	Unit $	Ext $	Supplier
7	C3	1	Bulk capacitor	VHT10M16	0.60	0.60	Vetco
8	J1	1	USB-C power breakout	Adafruit 4090	2.95	2.95	Adafruit
9	H1	1	Breakaway male header strip	Pololu 965	0.95	0.95	Pololu
10	W1	1	Jumper wire set	Adafruit 1954	1.95	1.95	Adafruit
				Total		**60.99**	**USD**

Supplier URLs

U1: Arduino Store — Arduino Nano Every

U2: Pololu — VL53L1X ToF sensor carrier

DS1: Adafruit — 0.96 in OLED display

S1: Vetco — Momentary pushbutton

R1: Estimated — Pull-up resistor

C1,C2: Futurlec/estimate — Ceramic decoupling capacitor

C3: Vetco — Bulk capacitor

J1: Adafruit — USB-C power breakout

H1: Pololu — Breakaway male header strip

W1: Adafruit — Jumper wire set

Firmware package

Included Arduino sketch: firmware/rangefinder_nano_every/rangefinder_nano_every.ino

Behavior: initialize OLED and VL53L1X, debounce button on D2, perform one ranging transaction per button press, and display either the measured distance or an invalid-target message.

Schematic QA summary

- All electrically relevant installed parts are shown: MCU, sensor, display, USB power input, button, pull-up resistor, and decoupling capacitors.
- All shown pins are labeled.
- Power and ground rails are explicit.
- Wires are drawn with orthogonal routing only.
- No wire crosses a component body in the rendered layout.
- The firmware pin map matches the schematic labels (A4 SDA, A5 SCL, D2 button).

Limitations

This is a concept-to-prototype package. It does not include enclosure CAD, formal PCB layout, EMC/ESD qualification, battery charging, or production DFM review. Prices are current/estimated from public listings and were not validated for stock at time of final packaging.